

Silvanus Brian Mulu

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Education

Texas State University

B.A. Economics (Honors) & Minor in Mathematics

• GPA: 4.0

San Marcos, TX

May 2029

Experience

Texas State University, Department of Economics & Finance

Undergraduate Research Assistant

Technologies: R, MATLAB, LaTeX

San Marcos, TX

Jan 2026 – Present

[\[🔗 Code\]](#) | [\[Work-In-Progress\]](#)

- Building a structural **input-output VAR (SVAR)** in R and MATLAB to trace **oil price shock transmission** through US industry linkages; decomposing **spot vs. speculative shocks** and estimating **impulse response functions** to quantify spillovers onto inflation, output, and the trade balance.

National Social Security Fund

Data Analyst Intern

Technologies: Power BI, Excel, Python

Nairobi, Kenya

Mar 2025 – Jun 2025

- Built **Power BI** dashboards to monitor payment flows and portfolio profitability across the national pension fund portfolio, enabling real time performance tracking
- Performed **pivot table analysis** on financial statements and reports to identify trends in fixed income, equities and real estate assets
- Collaborated with backend engineers on **schema design** and **data validation** rules for the member registration system

Projects

Statistical Arbitrage: Pairs Trading in U.S. Oil Equities

[\[🔗 Code\]](#) | [\[📄 Paper\]](#)

Technologies: R, Python, LaTeX

- Building a **statistical arbitrage research pipeline** in R, replicating and extending a Python-based study on **cointegration-driven pairs trading** across U.S. oil and energy equities.
- Implementing **Engle-Granger** and **Johansen cointegration tests** for pair selection, with **Ornstein-Uhlenbeck** half-life estimation to calibrate mean-reversion speed and holding horizons.

Newton Forward Method of Interpolation

[\[🔗 Code\]](#) | [\[📄 Paper\]](#)

Technologies: JavaScript, Node.js

- Implemented **Newton's forward-difference interpolation** from scratch using recursive divided differences an exploration on piecewise polynomial fitting up to **30th order**.

Leadership & Extracurriculars

International Student Organisation

Treasurer, Head of Finance Committee, Board of Director

San Marcos, TX

Sept 2024 – Present

- Oversee annual budget and financial planning for a 1,500+ member organization, ensuring fiscal accountability and strategic fund allocation.

Financial Management Association

Active Member, Capital Markets Cohort

San Marcos, TX

Feb 2025 – Present

- Manage a mock equity portfolio, conducting fundamental stock and sector analysis to inform allocation decisions.

Table Tennis Club

Active Member

Mbabane, eSwatini

Jan 2022 – 2024

- Bronze medallist at a collegiate tournament in eSwatini; 6-time gold medallist at high school tournaments in Kenya.

Awards

Shelby Davis Scholar: Full tuition, Texas State

President's List (2X): Perfect GPA

Goldman Sachs: Possibilities Series Participant (2026)

Beta Gamma Sigma: Invited Member, Honor Society (Top 0.5)

MIT Policy Hackathon: Globalist Finalist (2024)

Jump Trading Probability Cup: Top 5% algorithm forecast (2026)

Skills & Interests

- **Certifications:** Securities Industry Essential (SIE), FINRA - in progress
- **Research Interests:** Time Series Econometrics, Energy Economics, Empirical Macroeconomics, Machine Learning, Derivatives & Market Microstructure
- **Programming & Tools:** Python, R, SQL, MATLAB, JavaScript, Ruby; Excel (Advanced), Power BI, GitHub, Linux.
- **Languages:** English (Native), Swahili (Native), Kikamba (Native), Turkish (A2 – Beginner).
- **Interests:** Long-distance running, Poker, Chess, Checkers, Sudoku, Golf, Farming (300 chickens).